

vascular and endothelial function, coagulation pathways, blood pressure, glucose homeostasis, insulin sensitivity, and body fat distribution.^{4,5} These pathways also undergo unfavorable changes during the menopause transition and postmenopause, contributing to increasing cardiometabolic risk with aging.^{2,6} Kidney health is also closely linked to menopause and estrogen status. Ovarian senescence and the loss of estrogen's potentially renoprotective effects may contribute directly to declining kidney function and indirectly through the accumulation of cardiometabolic risk factors that predispose women to chronic kidney disease. Although estrogen may influence renal function, epidemiologic findings regarding the kidney effects of HRT remain conflicting, and the optimal use and dosing of hormone therapy among women with impaired kidney function have not been established. These uncertainties are particularly relevant to older postmenopausal women, in whom age-related kidney function decline and comorbidities such as hypertension, diabetes, and dyslipidemia are increasingly common.⁷

Much of the ongoing concern about HRT safety arises from RCTs showing that its effects differ according to regimen and patient characteristics. In the Women's Health Initiative Hormone Therapy Trials (WHI-HT), combined estrogen-progestin therapy was associated with increased risks of coronary heart disease, stroke, venous thromboembolism, pulmonary embolism, and breast cancer.⁸ By contrast, estrogen-only therapy remained associated with increased risks of stroke and venous thromboembolism but did not demonstrate the same increase in breast cancer seen with combined estrogen-progestin therapy.⁹ Subsequent age- and timing-stratified analyses resulted in the timing hypothesis, which proposes that the cardiovascular effects of HRT vary based on the interval between menopause onset and treatment initiation.^{10,11} Accordingly, the North American Menopause Society (NAMS) recommends that the benefit-risk profile of HRT is generally favorable for women younger than 60 years or within 10 years of menopause onset.¹

Consistent with this evolving evidence base, in February 2026, the US Food and Drug Administration (USFDA) approved labeling changes for six menopausal HRT products that removed boxed-warning statements concerning cardiovascular disease, breast cancer, and probable dementia, while retaining the boxed warning on endometrial cancer for systemic estrogen-alone products.¹²

Despite substantial progress in understanding HRT effects in menopausal women, important uncertainty remains for women who initiate therapy at older ages, particularly after age 65 years or more than 10 years after menopause onset.¹ This population is especially important because aging is accompanied by greater subclinical atherosclerotic burden, endothelial dysfunction,^{13–15} and longer cumulative exposure to estrogen deficiency, all of which may alter the physiologic and clinical effects of HRT. At the same time, older women remain underrepresented in contemporary evidence syntheses focused specifically on cardiometabolic outcomes, leaving clinicians with limited data to guide decisions regarding late initiation or continuation of therapy.

A focused synthesis of evidence in this population is therefore needed to improve individualized benefit-risk assessment. Although two recent narrative reviews^{4,5} summarized the broader evidence on HRT and cardiometabolic or cardiovascular health, neither systematically focused on randomized and post-trial evidence relevant to women aged 65 years and older or included renal outcomes as a predefined outcome domain. Accordingly, this systematic review aimed to identify, critically appraise, and synthesize evidence evaluating the cardiovascular-kidney-metabolic effects of HRT in postmenopausal women, with particular emphasis on women aged 65 years and older. The review also assessed the strengths and limitations of the available evidence, including risk of bias, identified important knowledge gaps, and highlighted priorities for future research, regulatory evaluation, guideline development, and personalized clinical decision-making.

Methods

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guideline.¹⁶ The protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO)(CRD420261338310).

Eligibility Criteria

Eligibility criteria were defined according to the PECO framework and study design. The population (P) of interest was postmenopausal women, with particular emphasis on women aged 65 years and older. Studies conducted in broader postmenopausal populations were also eligible if they reported subgroup, stratified, or follow-up findings relevant to